Block Caving

13/05/2020

Principle I

*Block caving refer to a mass mining system where extraction of ore largely depends on action of gravity.

*A thin horizontal layer mining level of ore column is removed, using standard mining method.

*This removes vertical support of ore column above, then ore caves by gravity.

*

Principle II

*Broken ore of ore column is removed from mining level.

*Ore above continues to break and cave by gravity.

*Block caving is cheapest under ground method of mining. Block caving method does not require explosive to break rock for production.

*Block caving method does not require drilling & blasting to breal rock for production.

Geological exploration through bore holes

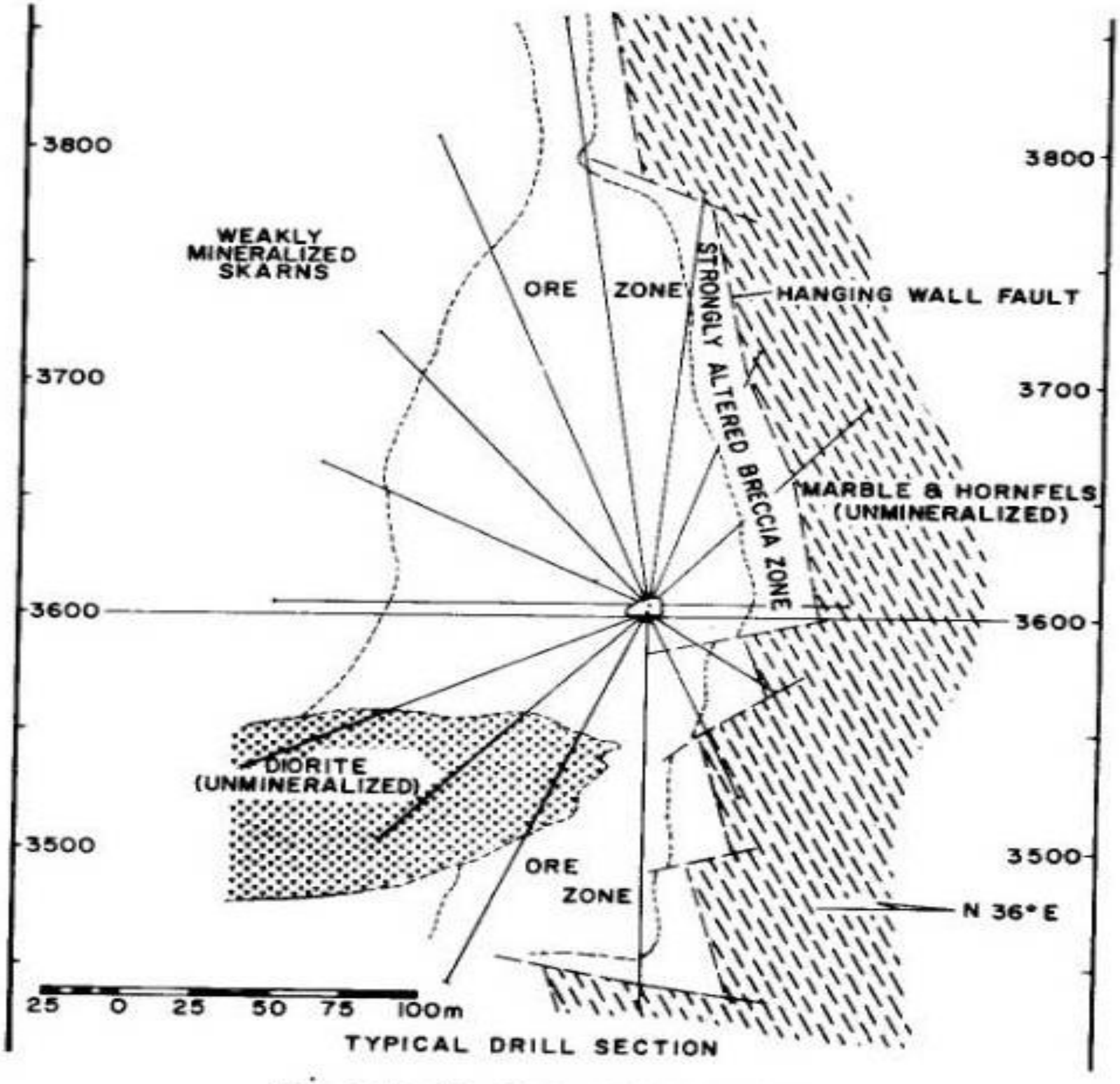


Fig. 20.3.17. Typical drill section.

Applicability I

*Ore body should be of porphyry type with well disseminated mineralization and fairly large lateral & vertical extend.

*It can be applied to steeply dipping deposit or flat laying deposit of sufficient width & thickness.

*Rock strength can be fairly weak or fairly strong.

*Total mass must have sufficient fractures in different orientations to allow rock mass to break up under gravity in to pieces small enough to pass through draw wholes in to production drift.

*Lateral extend of ore body must be large enough to ensure that a cave can be established.

Applicability II

*Horizontal area required to establish a cave will depends on strength of rock mass.

*Minimum horizontal dimension of mining area should be about 90 m sq..

*Height of column should be sufficient to allow a reasonable productive life to individual draw points.

*Height of column shall also ensure reasonable rate of return on development and production cost.

*Rate of return also depends on value of mineral in ore.

Determining Cavability I

*Cavability of deposit must be determined at beginning of stope design.

*Frequency and orientation of various fracture sets must be evaluated.

*Rock Quality Designation (RQD) of drill core is an indication of cavability.

*RQD is determined by measuring length of recovered drill core for 100 drill length.

*Usually, incremental value for drill length is 3m.

*RQD does not give fracture orientation, intensity, fracture set etc.

Determining Cavability II

*Access to ore body is essential, before a final determination of cavability can be made.

*At least 3 sets of major fractures are necessary to justify use of block caving.

*Two vertical sets at approximately 90 degree orientation to each other & third set at approximately horizontal, will form rectangular block when fracture separates.

*Spacing of fractures is also important since spacing will determine size of block that will appear at draw points.

Determining Cavability III

*Fracture and joint fill material must also be identified, since composition of that material will help to determine how easily fracture will separate or if fracture fill material is actually stronger than host rock.

*Similar study is to be made for waste capping to be sure that it will also cave as ore is removed.

*In case capping breaks finer than ore, then dilution will be a greater problem with greater amount of waste infiltrating in to ore.

*Major faults & dikes must be identified, as they also effect caving action and on mine layout.

Determining Cavability IV

*Rock Mass Classification (RMC) incorporates such information as joint spacing, joint conditions, intact rock strength, joint filling & other factor.

*Rock mass classification indicates ease or difficulties in cavability.

*Cavability of ore block depends on pre splitting (boundary loosening) of orebody.

*Cavability is a factor of cementation in fracture & joint plane.

Production Capacity I

*Block caving is best suited for high daily production.

*Up to 80,000 ton per day production can be achieved by block caving.

*Production capacity of a mine is determined by rate of vertical draw that can be made and horizontal area that can be under cut at any given time with out creating ground control problem.

*Draw rate varies from 150mm to 600mm per day.

*Daily rate of draw will be determined by rate at which ore will cave.

*Strong, less fractured rock likely to cave very slowly.

Production Capacity II

*Weak, highly fractured rock likely to cave rapidly.

*There is no method to predetermine, how rapidly a given ore body will cave.

*Experience at other mine with similar type of rock will provide a guide line.

*Production capacity of stope depends on frequency & efficiency of secondary blasting at draw point.

*Number of available draw point for production effects production from stope.

Production Capacity III

*Large number of damaged or non furnacing draw point shall reduce production capacity of stope also increase dilution.

*Equipment capacity & availability determines production.

*Nature of hang wall & foot wall effects production from block caving.

Development I

*Rate of development must balance with production rate.

*When number of potential ton added through development is equal to ton drawn, productive capacity remains constant.

*If tonnage drawn varies significantly, some adjustment has to be made in development program, to maintain production potential.

*Amount of lead time required to prepare draw point must be calculated carefully.

Development II

*Lead time required will depends on available man power & efficiency of man power & equipment being used.

*About 3 years will be required to prepare a new production drift from time of initial access is started until under cut long holes are blasted.

*Undercut drift takes about 6 months to 1 year , before blasting to allow adequate time to complete long holing.

*Long holing should precede blasting by about 6 months.

Draw Point Spacing I

*Draw point spacing is important to insure good recovery of ore.

*Draw point spacing depends on material size.

*When ore breaks very fine, draw points will be closer.

*For course material, draw points shall be spaced wider.

*Influence of one draw point is determined by size of material in cave.

*For fine ore, area of influence is small & for course ore , area of influence is high.

Draw Point Spacing II

*Draw point pacing is too far, then good ore shall be lost & dilution mat become too high making mining uneconomical.

*If draw points are too close, mining cost goes up substantially.

*Each draw point has certain area of influence on broken ore above.

*Draw point spacing should be so spaced that these areas of influence will over lap slightly to ensure total ore column is moving downward as draw progress.

Draw point Number I

*Number of draw points required for a given level of production depends on

(1) rate that ore will cave in per day

(2)amount of tonnage that will flow through draw point

(3)productive capacity of equipment as provided by manufacturer with practical allowances.

*Very weak ore normally caves rapidly, where stronger ore caves at very slow rate.

Draw point Number II

*Rate of draw should be fast enough, so that there is room for ore above to cave but not so rapidly that a large void is created between in-place ore & broken ore.

*Draw rates can vary any where from 150 to 610mm per day.

*Amount of tonnage that can be expected from a given draw point also depends on how many hang-up occurs during a given time period.

*Once number of draw points required has been determined, a small percentage should be added to allow for hang-up & for repair.

Draw point Number III

*Availability of new draw point is in between 80 to 90%.

* Availability of old draw point comes down due to repair etc.

*Availability of production equipment should also include haulage equipments for calculating potential production capacity.

*All loading points should not be concentrated in one or two loading drift.

Selecting Mining System I

*Size of broken material will decide which mining system is most suitable.

*Mine site location, availability of skill work force, labour cost, availability, maintainability and cost of equipment, capital fund also influence system selection.

*Draw point spacing effects selection of method of sublevel caving.

*Draw point distance depends on material size.

*When ore breaks very fine, draw points will be closer.

Selecting Mining System II

*For course material, draw points shall be spaced wider.

*Influence of one draw point is determined by size of material in cave.

*For fine ore, area of influence is small & for course ore , area of influence is high.

*

LHD System IA

*LHD system is mechanized method of working & possible to adopted wide range of ore body with some modification.

*When geologic studies shows that ore will break in to relatively larger piece, LHD (rubber tired equipment) works much better than other system of block caving.

*

LHD System II

*LHD requires that draw points be separated at greater interval to allow rooms for LHD to operate.

*Course ore provides the possibility of wider gap in between draw points.

*System consists haulage level, ore transfer raises, production levels, draw points entries, draw cones to under cut and under cut drift.

LHD System III

*Long holing & blasting is used to form under cut that promotes caving.

*Large draw cone allows larger pieces of ore to move down near draw points.

*Small drill machines & explosives are used to break up larger pieces.

*Using separate access, haulage, production & under cut levels can be developed simultaneously.

*Haulage & production levels should be separated by a substantial distance to provide adequate storage un transfer raise, so that loading of train is not delayed, waiting for LHD.

LHD System IV

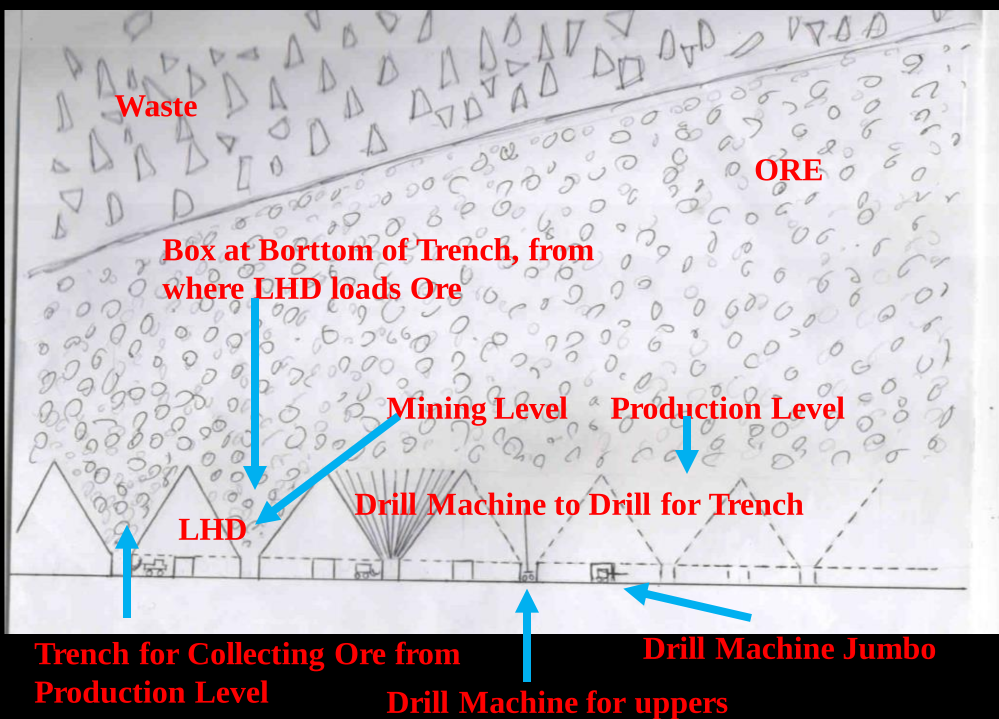
*After excavating haulage & production drift, transfer raise & draw points entries can be driven.

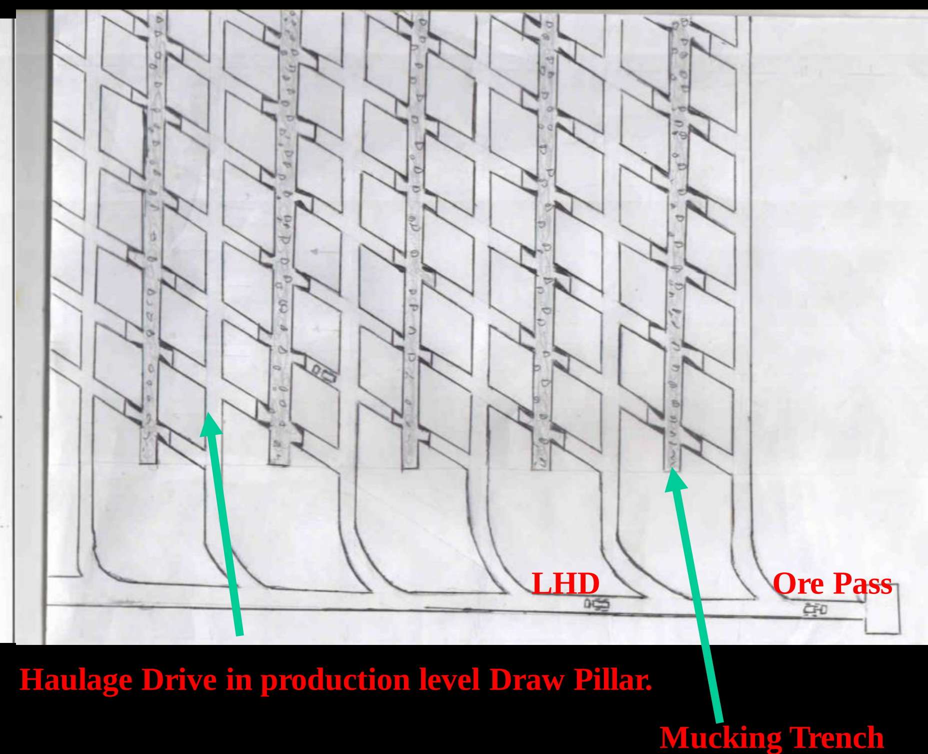
*Drilling of draw cones can be done from draw points entries or from under cut levels.

*It is safer to excavate draw cones from under cut level.

*Long holing from under cut is final stape before under cut is blasted.

Mechanised Block Caving In Trench* LHD Mucking





LHD system layout showing draw cone, production / haulage drift

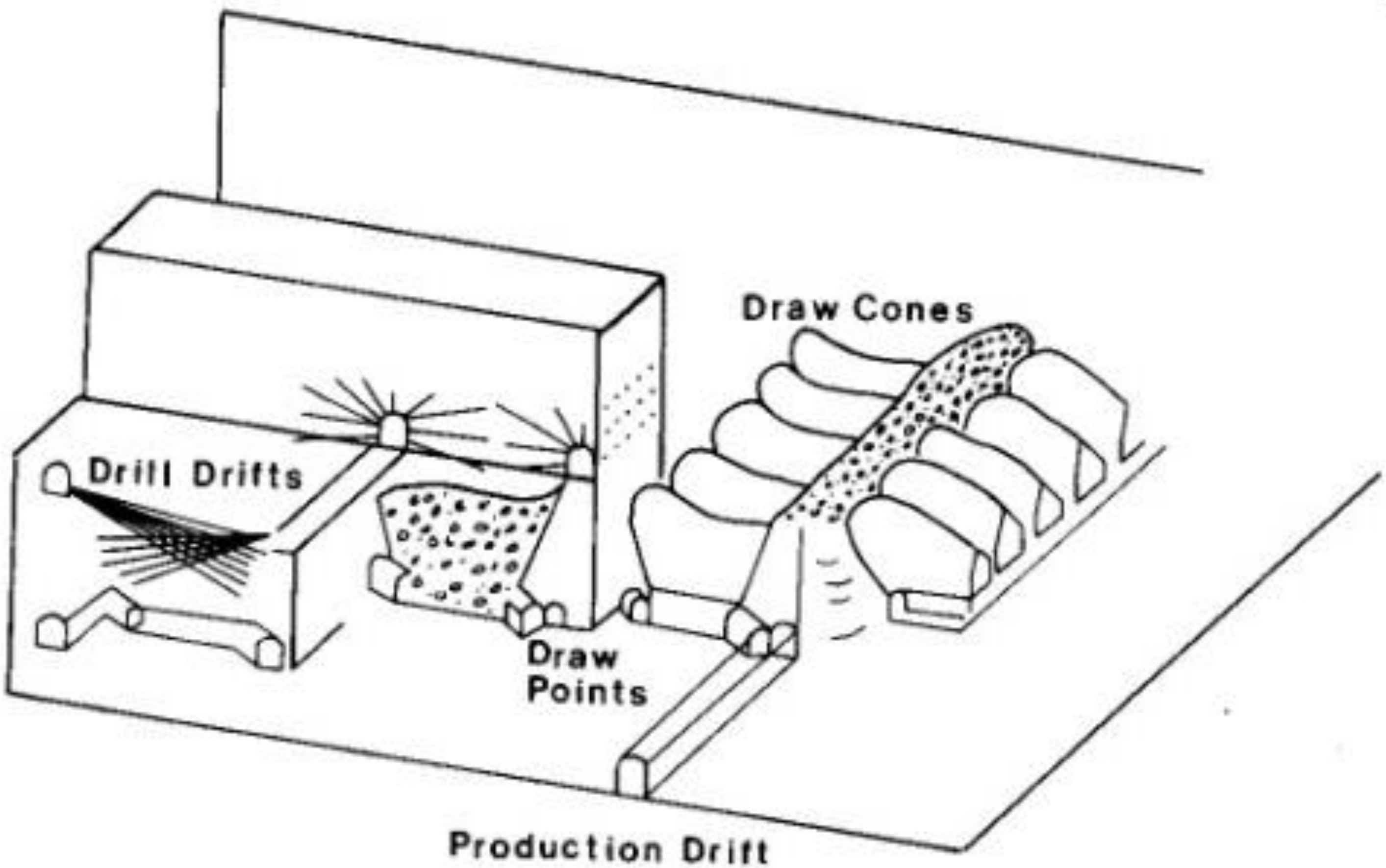


Fig. 20.3.3. LHD system—typical layout.

LHD draw point & Cone

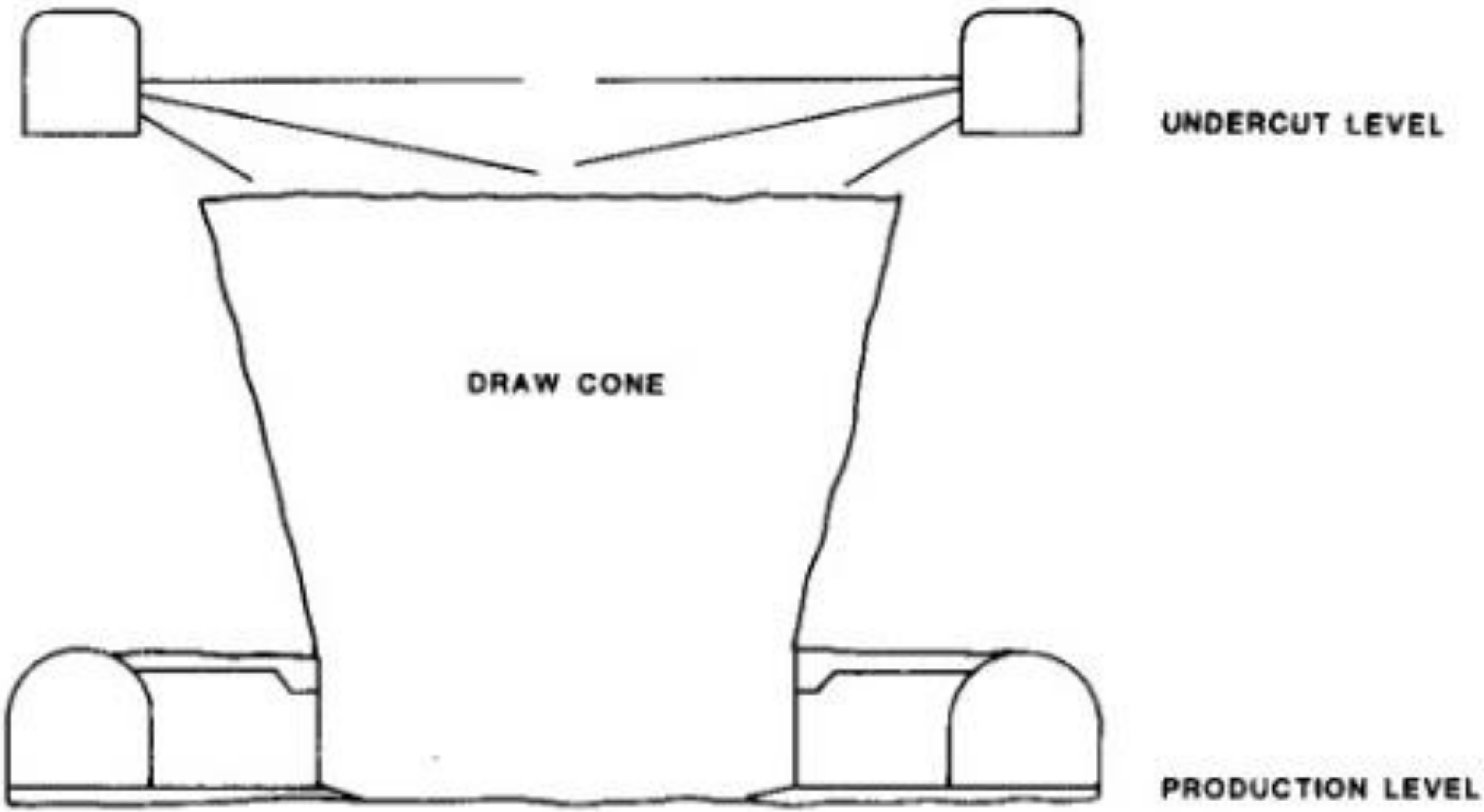
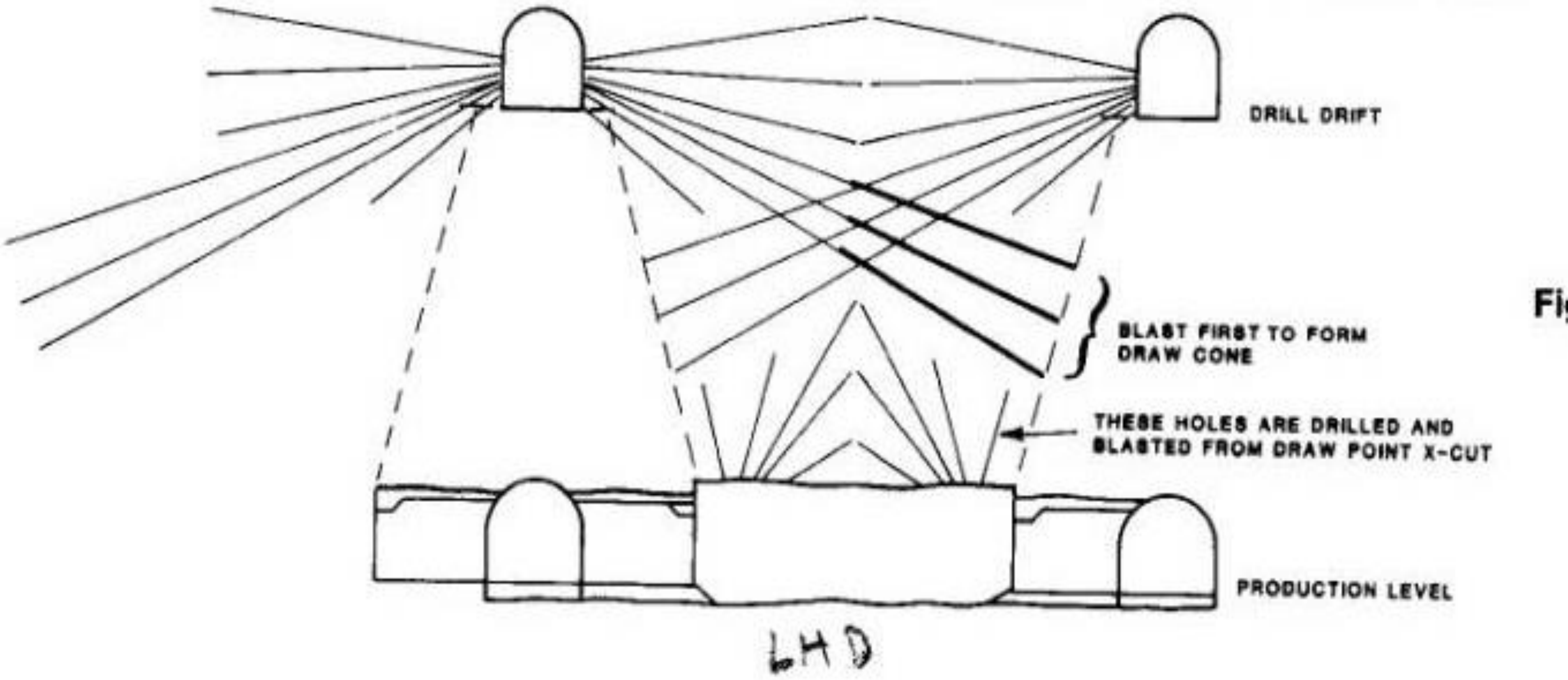
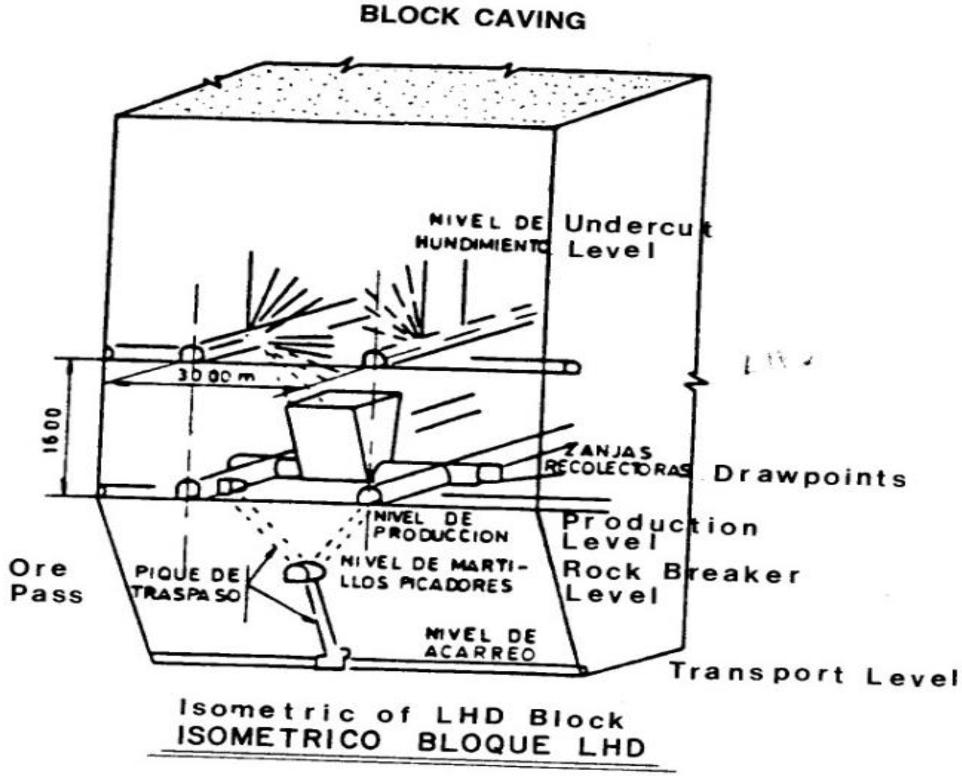


Fig. 20.3.9. LHD drawpoint and cone.

Drilling pattern for draw point & undercut



LHD mining system



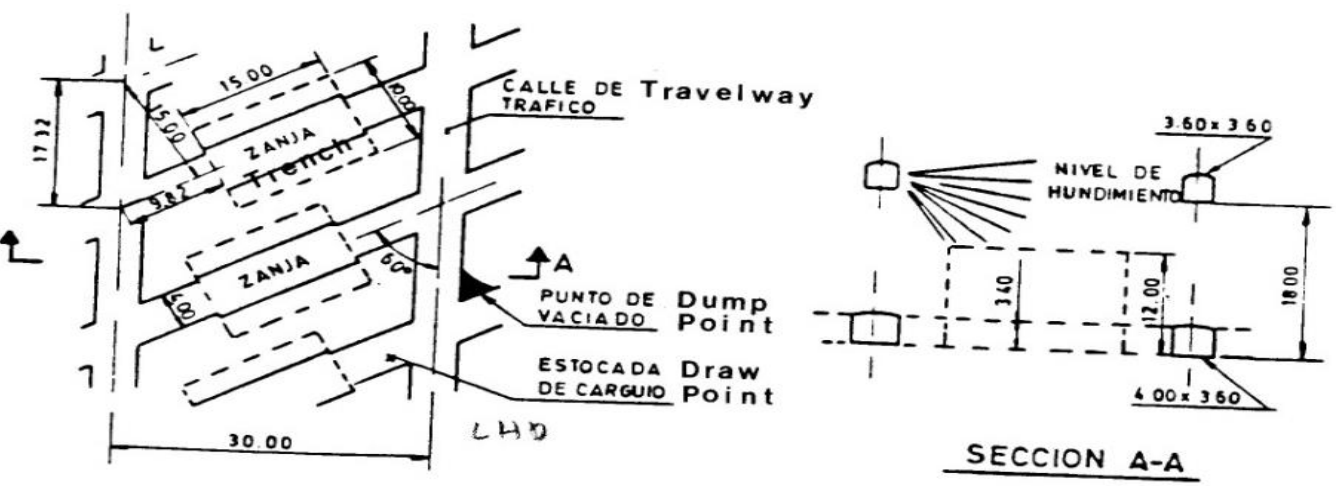


Fig. 20.3.12. Isometric of LHD block, plan of production block.
Conversion factor: 1 ft = 0.3048 m

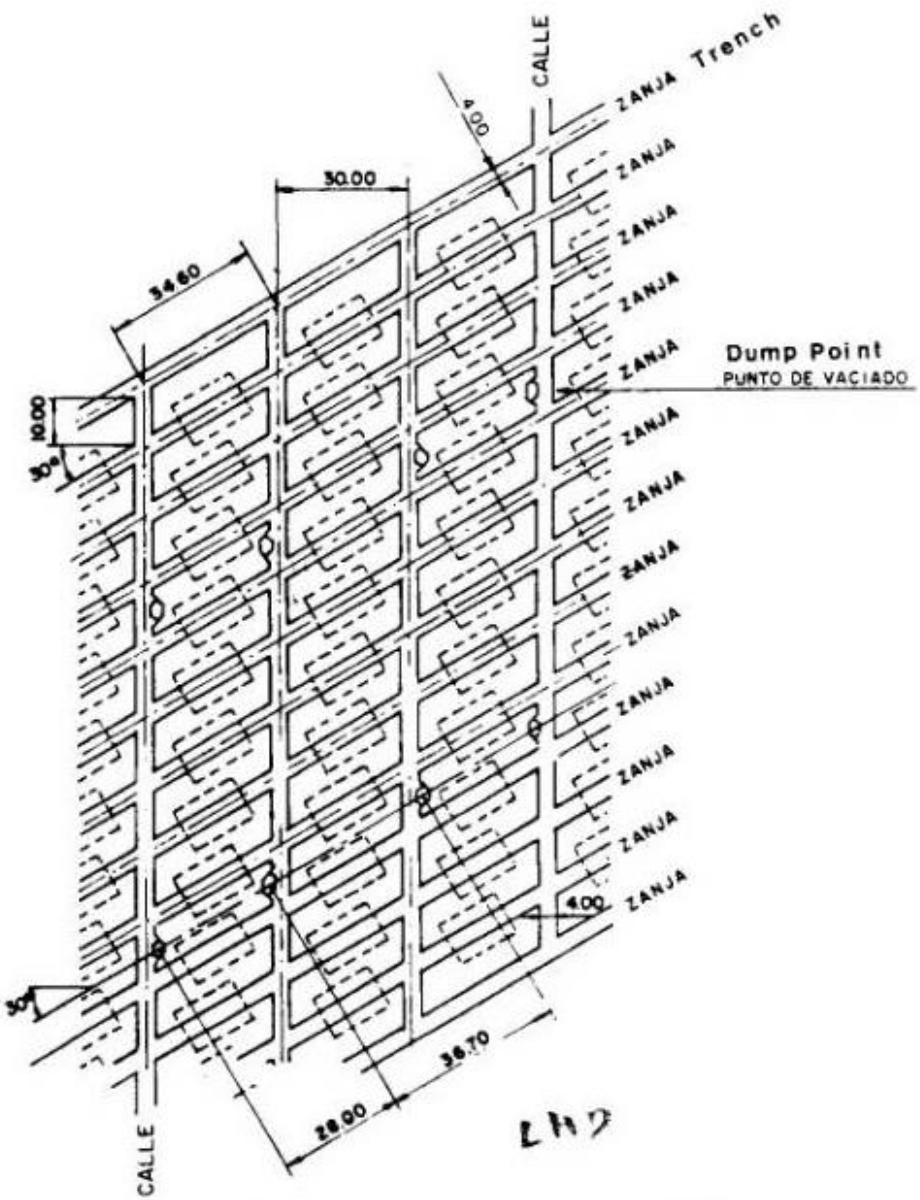


Fig. 20.3.13. Production level layout — El Teniente.
Conversion factor: 1 ft = 0.3048 m.

LHD system production level

Production level support LHD system

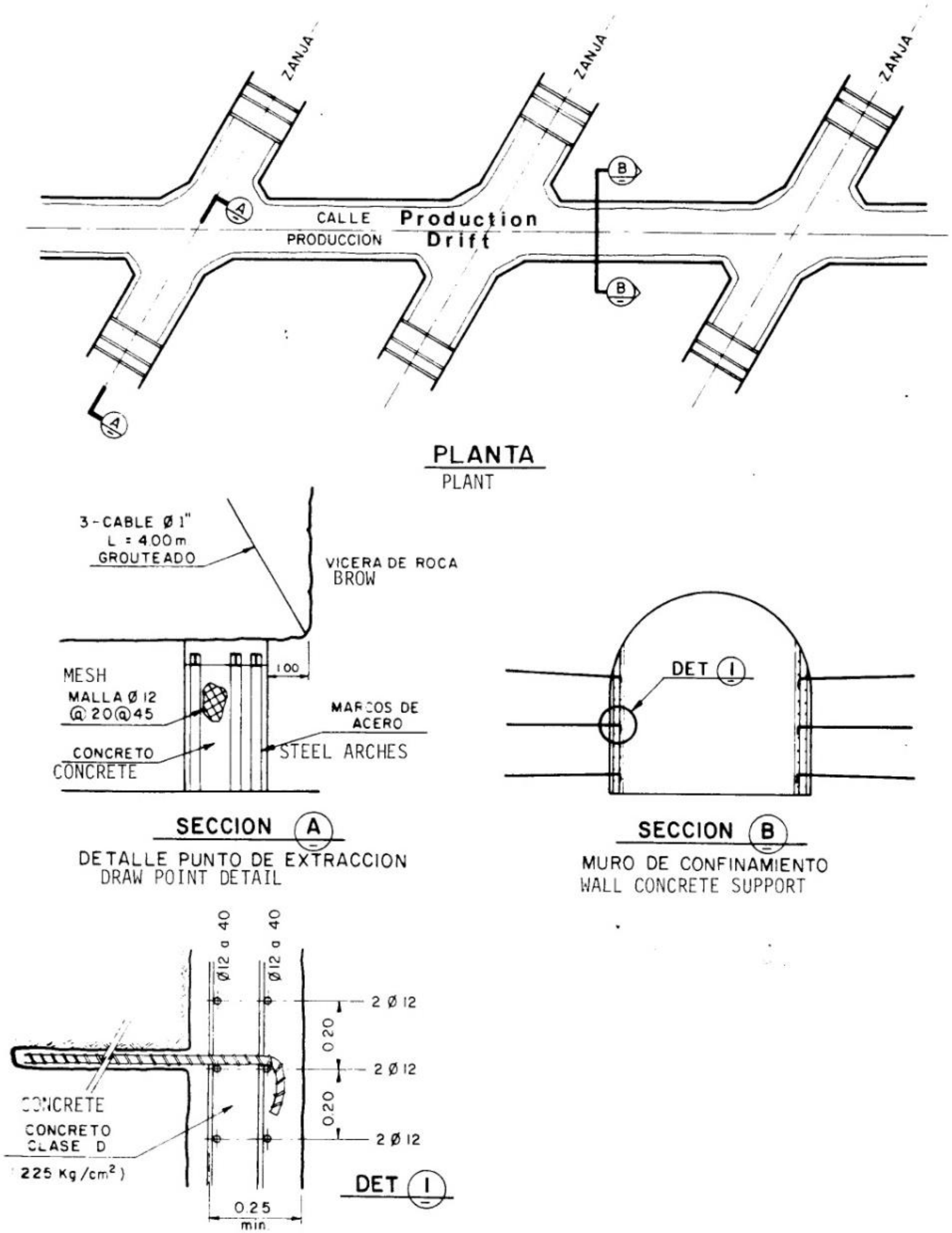


Fig. 20.3.14. Production level support.
Conversion factors: 1 in. = 25.4 mm, 1 ft = 0.3048 m, 1 psi = 6.895 kPa.

LHD

LHD system draw points

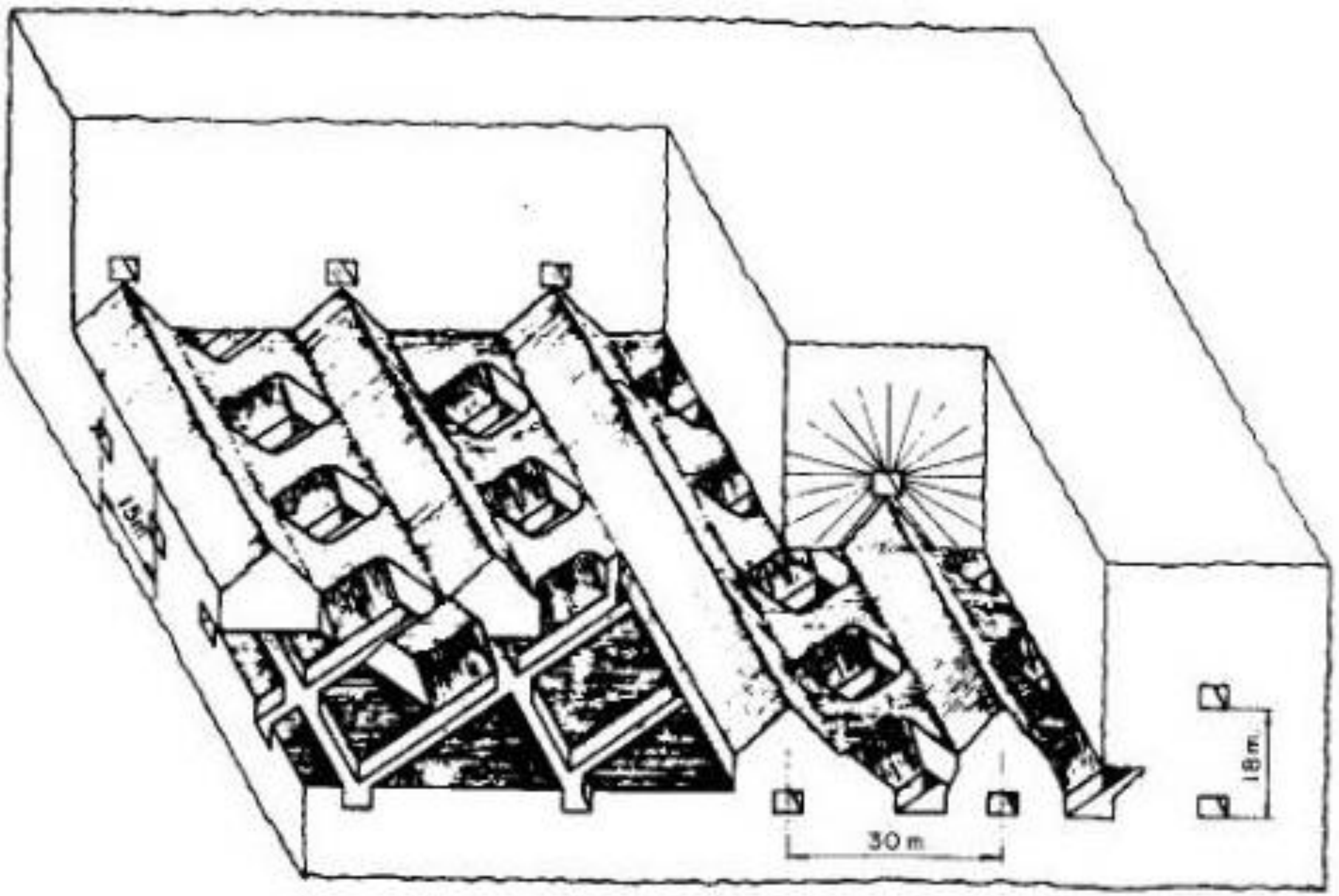
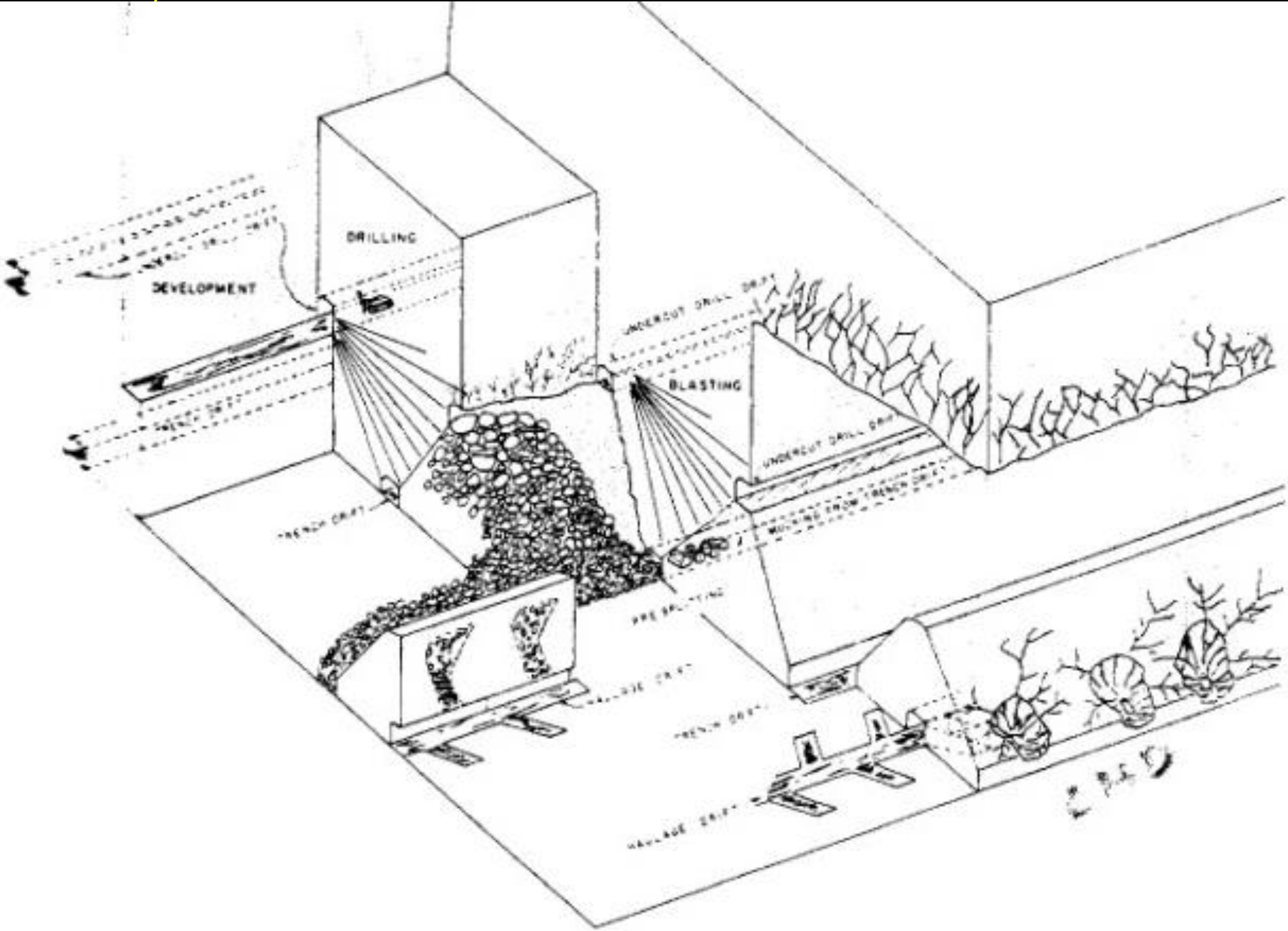


Fig. 20.3.16. Isometric of LHD system — El Teniente.
Conversion factor: 1 ft = 0.3048 m.

LHD system with trench down



Grizzly System I

*When ore breaks very fine, draw points will be closer.

*Grizzly system requires that ore to break at very fine size.

*Draw point spacing in Grizzly system is closest.

Grizzly System II

*Grizzly (Full Gravity) system consist of haulage level, transfer raise, grizzly level, finger raises & under cut level.

*Haulage & grizzly levels are driven across ore block to be mined at different elevation.

*This work can be done simultaneously since they are on different levels.

*Finger raises are driven from grizzly level to under cut elevation.

*Short horizontal connections are driven between top of finger raises to form pillar.

Grizzly System III

*Transfer raises are driven between haulage and grizzly level with loading chute at haulage level.

*Grizzly is provided with horizontal parallel rails / bars at specific interval to arrest big boulders.

*Under cut is blasted & broken ore starts flowing down finger raises.

*Some time large rock pieces hangs up in finger raises.

*Large boulder are broken by secondary blousing placing explosive strategically over size boulder..

*At grizzly ore is screened by rails for arresting big boulders.

Grizzly System IV

*Big boulders are broken by sledge hammer to smaller pieces.

*Some mine uses rock breaker in place of sledge hammer for breaking boulder at grizzly.

*Ore through transfer chute is loaded in to mine car with help of chute gate.

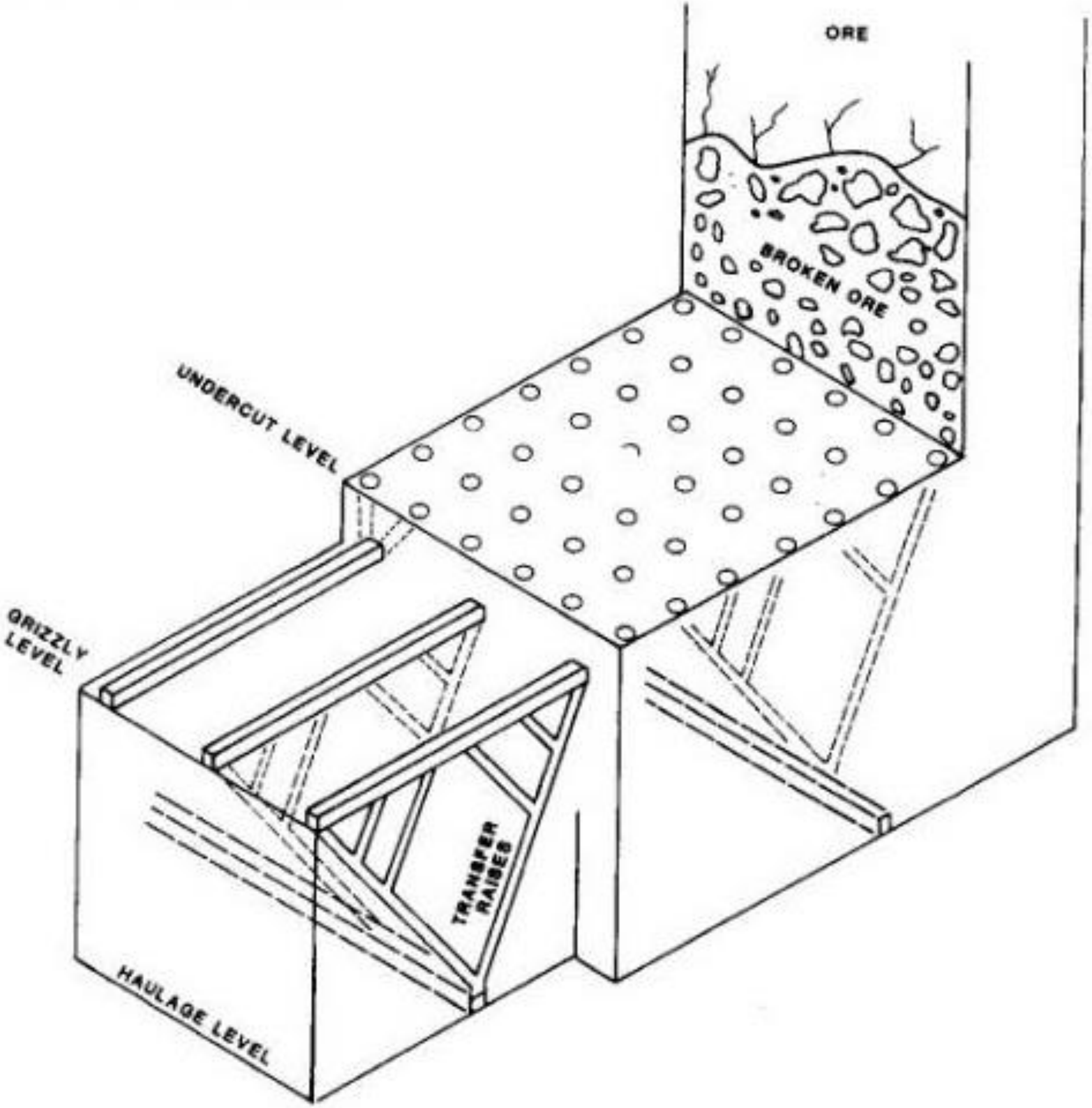
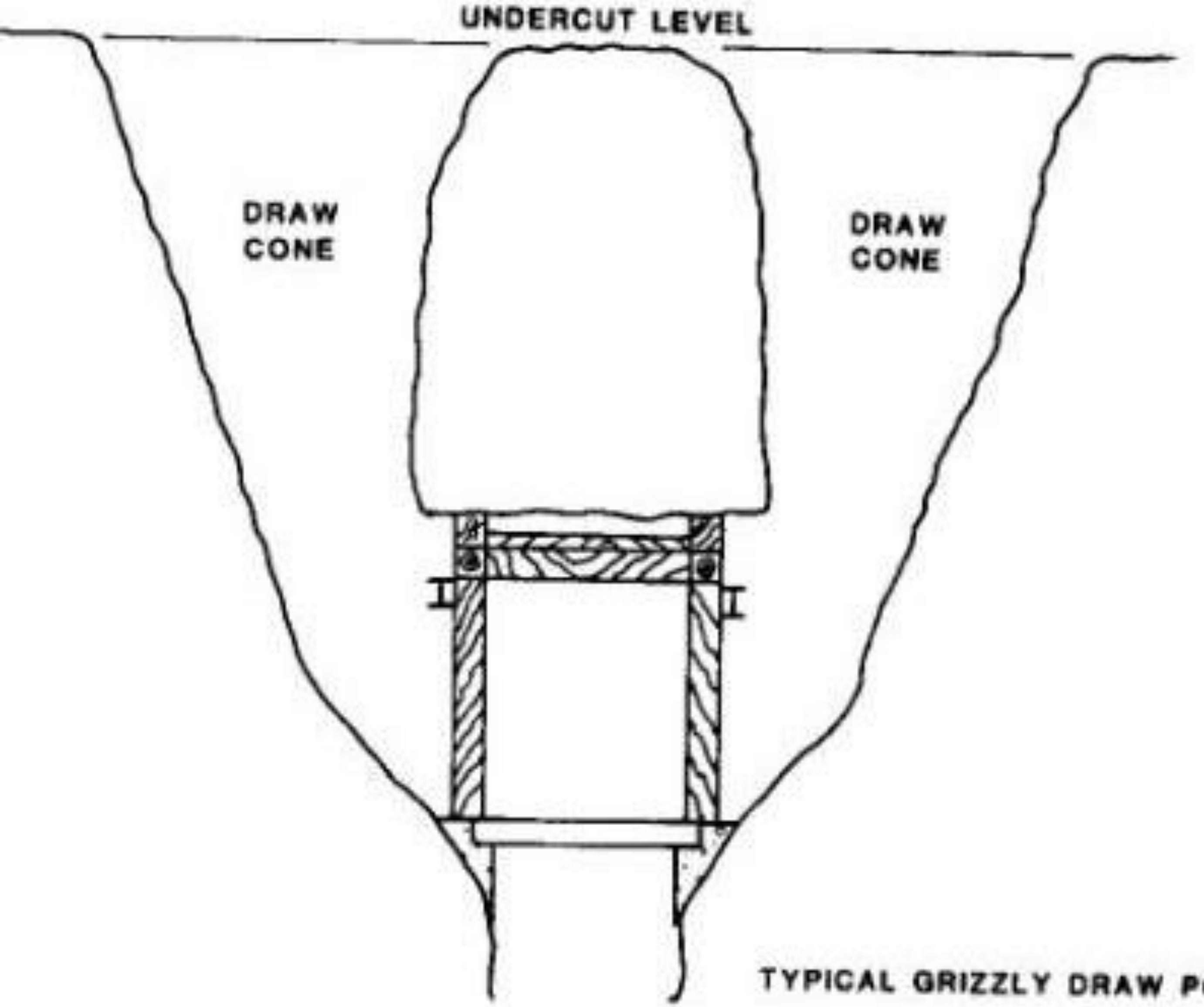


Fig. 20.3.1. Grizzly system—typical layout.

Grizzly System showing under cut, ore pass erc

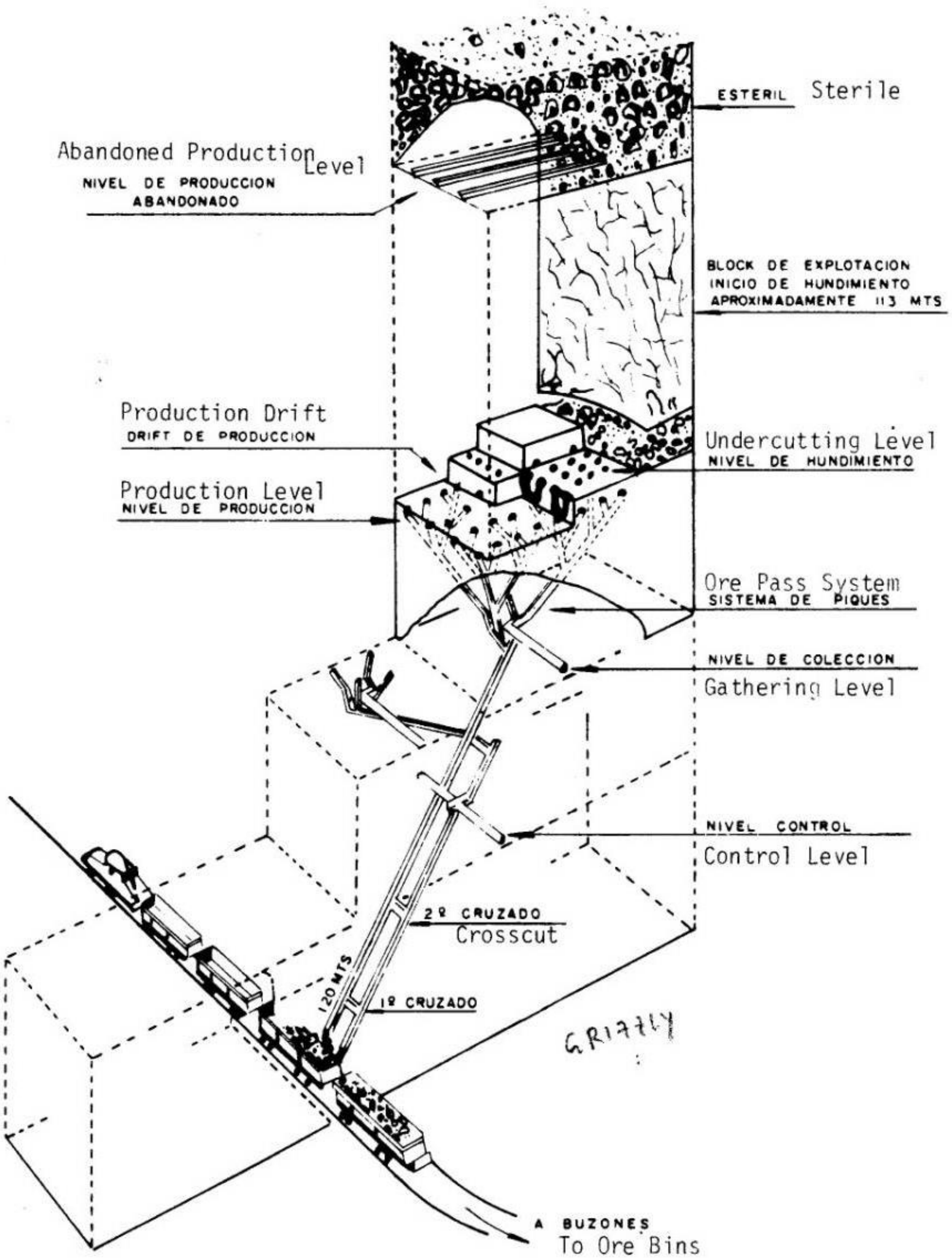


TYPICAL GRIZZLY DRAW POINT

her and grizzly.

Grizzly level draw point

Grizzly system



Slusher System I

*Slusher system is used for rock that will break in to moderate size pieces.

*In slusher system, draw points are placed at moderate space.

*Number of draw points can be in reach of one slusher travel distance.

*System consists of a haulage level with car loading arrangement.

*Finger raises are excavated connecting slusher level and under cut level.

*Haulage drift is driven along center of ore body block to be mined in strike direction.

Slusher System II

*Ventilation drift is driven at haulage level for ventilating slusher drift and haulage drift.

*Slusher drifts are at right angle to haulage drift.

*Slusher (scrapper) level is just above haulage level.

*Alternate slusher drifts are placed at 108 degree (mirror image) of other slusher drift above haulage drift.

*Slusher drift is supported by concrete.

*Finger raises are excavated to under cut elevation from slusher drift.

*Under cut drift is excavated from finger raises.

Slusher System III

*Under cut drift can also be developed simultaneously with lower level development through separate access.

*Long hole drilling is done from undercut drift.

*Long holes are blasted to complete under cut.

*Ore caving starts with flow of ore in to chute raise

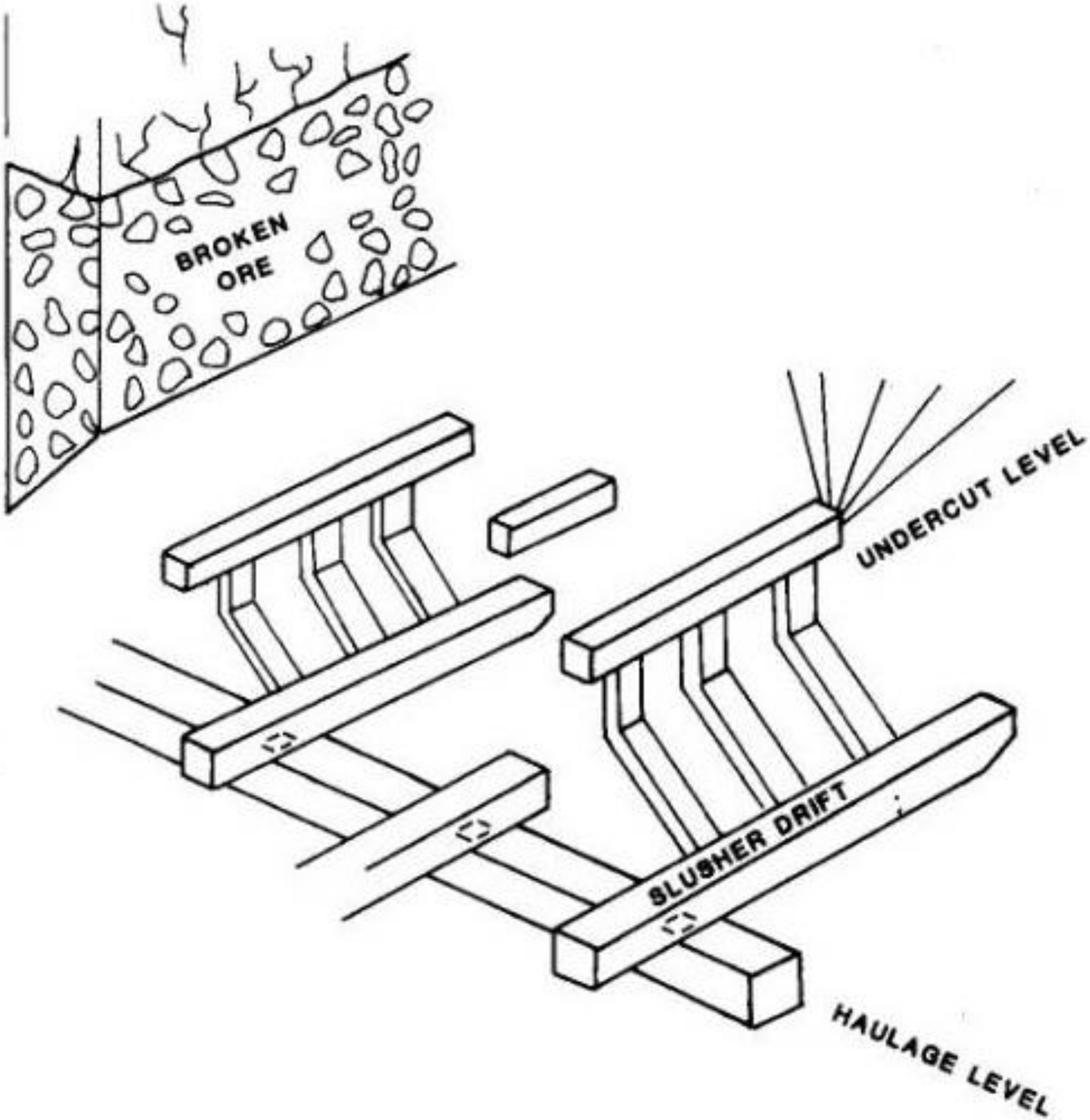


Fig. 20.3.2. Slusher system—typical layout.

Slusher system, showing under cut level, slusher drift, haulage level

Slusher level draw point on haulage level

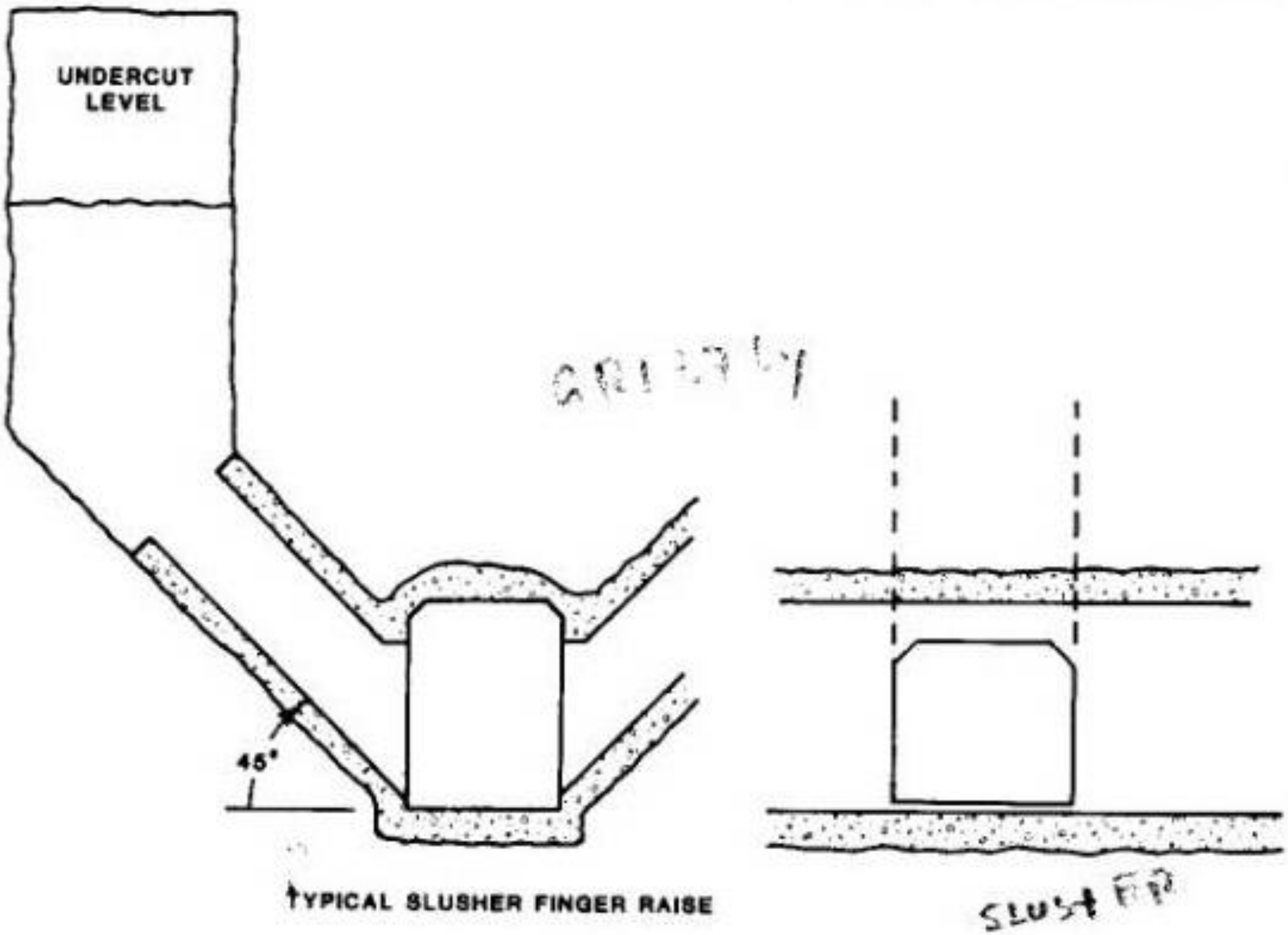


Fig. 20.3.4. Drawpoint design–slus

Slusher Drift & under cut drill

EERING HANDBOOK

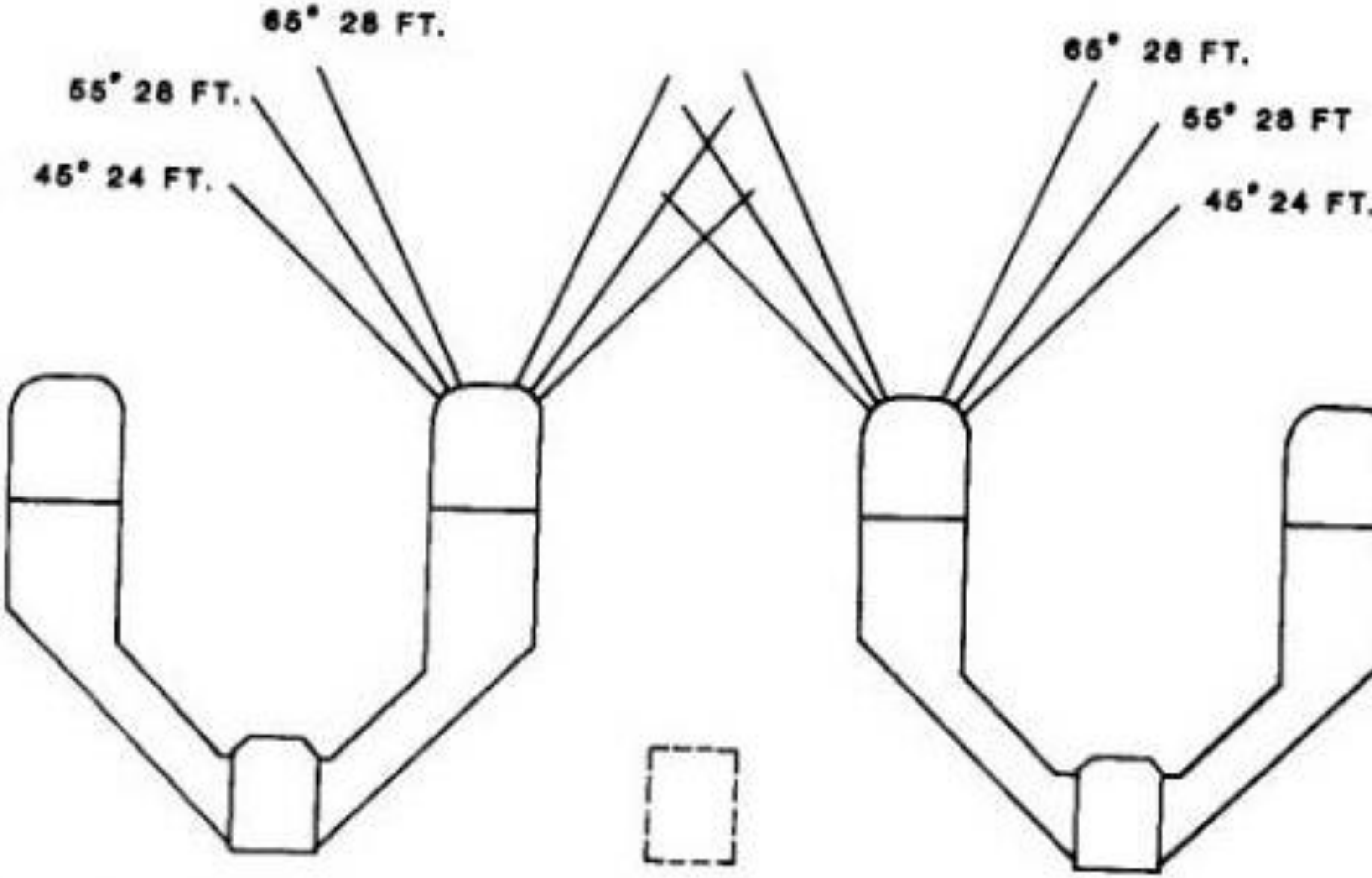


Fig. 20.3.8. Typical drill pattern for undercut blasting — slusher drift. Conversion factor: 1 ft = 0.3048 m.

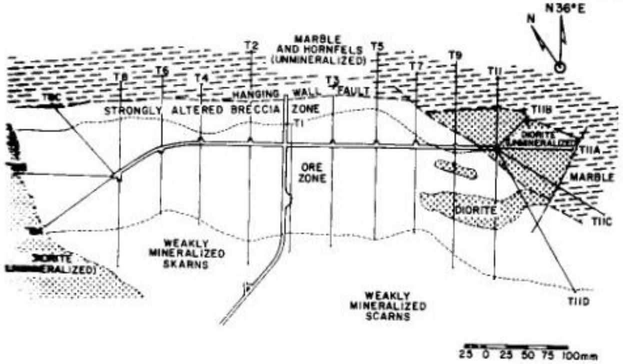


Fig. 20.3.18. Generalized 3600 level plan.

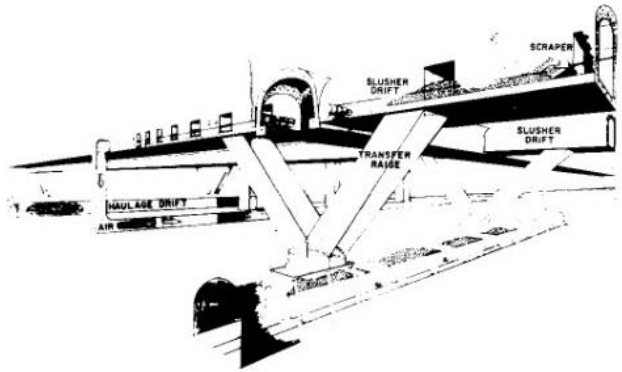
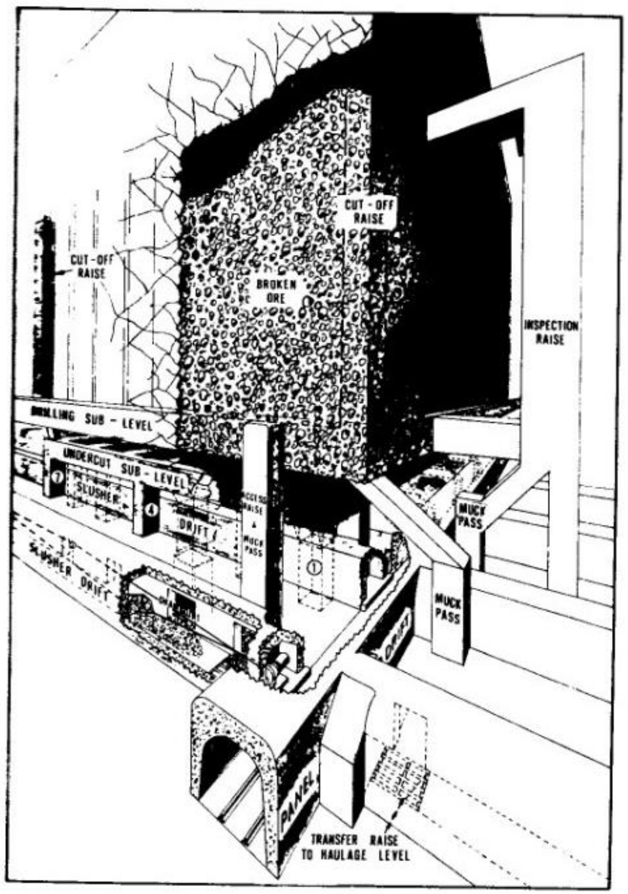
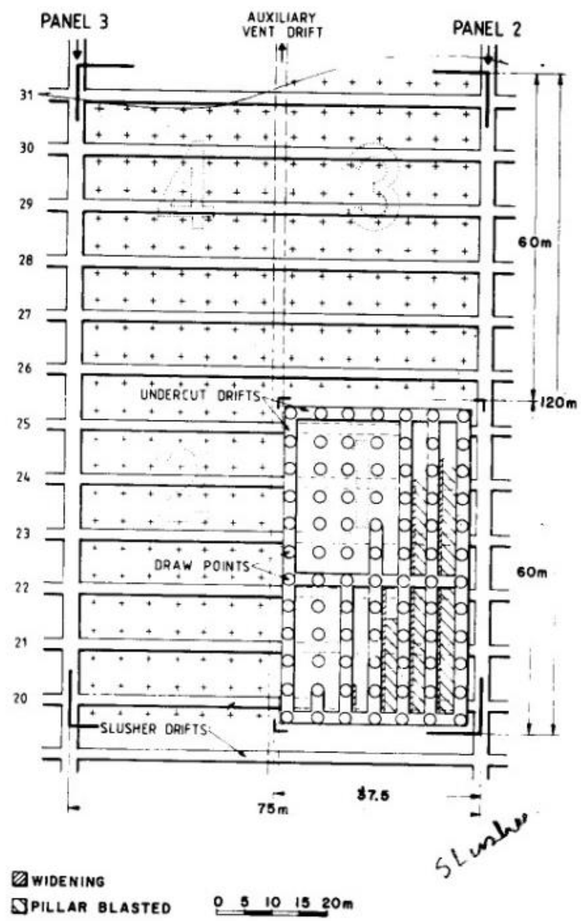


Fig. 20.3.20. Ore transfer raise system.





Slusher system mining with production level

Haulage Level I

*Haulage level is located under production level.

*At grizzly system, short transfer raises are excavated between production (grizzly) level & haulage level, allowing some storage of ore for loading to mine car.

*Capacity of ore pass is so decided, so that there is no interruption to mine car loading.

*Length of ore pass can be increased to allow more ore storage space.

*In slusher system, haulage level is usually beneath slusher loading hole, so that ore can be loaded directly in to haulage car.

Haulage Level II

*Some time transfer raises are used in between slusher level and haulage level.

*In LHD system, haulage & production levels should be saperate by large distance to provide adequate storage of ore.

*This distance also helps to reduce number of raise required to meet production.

*Raises should be of 60 to 70 degree from horizontal.

*This ensures good flow of ore and provide additional breakage of larger rock fragment as they drop down ore pass.

Haulage Level III

*Haulage level is excavated below grizzly (production) level for hauling ore out of system.

*Distance between haulage level & grizzly level is decided based on the ore to be held up in transfer raise for smooth loading to cars.

*This distance is about 18m

*Longer ore pass is used when ore is to be transferred to central gathering level.

*In case of long ore pass, there should be a short section just below grizzly level so that ton of ore drawn can be measured for draw control purpose.

Haulage Level IV

*Haulage level is subjected to extreme wet conditions,

*Haulage drift are supported with concrete.

*Cross section of haulage way is decided by haulage equipment.

Production Level I

*In grizzly system, grizzly level becomes production level.

*Production levels are of fairly small cross section.

*Production levels are used for access to finger raises and grizzly for sizing ore before it is sent down transfer raises.

*It is desirable, that production level are to be supported (lined) by concrete.

*Slusher system, slusher drift is production level.

*Production drift should be large enough to accommodate slusher & hoe.

Production Level II

*Scrapper usually requires 1.5m to 1.8m width drift.

*Production drift should be lined with concrete for support.

*LHD system, drift which provide access to LHD is called production drift.

Undercut Level I

*Under cut drifts are taken at suitable distance over top of draw points.

*Under cut level is used for long hole drilling, starting final under cut blast to initiate caving.

*Under cut should be at proper alignment with woking below.

*Cross cut are excavated between under cut drifts to facilate access between drifts.

*Cross cut also delimit area to be blasted during some specific time period.

Long Hole Drilling I

*Long hole drilling is step before under cut blasting.

*Purpose of long hole blasting is to remove pillars between pillars between under cut drifts.

*This blast forms a horizontal slots, that will allow ore above to commence caving.

*Long holes are to be drilled horizontally through pillars or at some angle above under cut level to form an apex between draw holes.

*Draw holes are flattered at angle usually 45 degree.

Long Hole Drilling II

*Drilling pattern should be spaced evenly along under cut drift with spacing between drill pattern determined by hole size and explosive load to be used.

*Height of under cut can very from 2.5 m to 15m.

*In case of bigger block ore, caving shall be difficult to initiate with short height under cut.

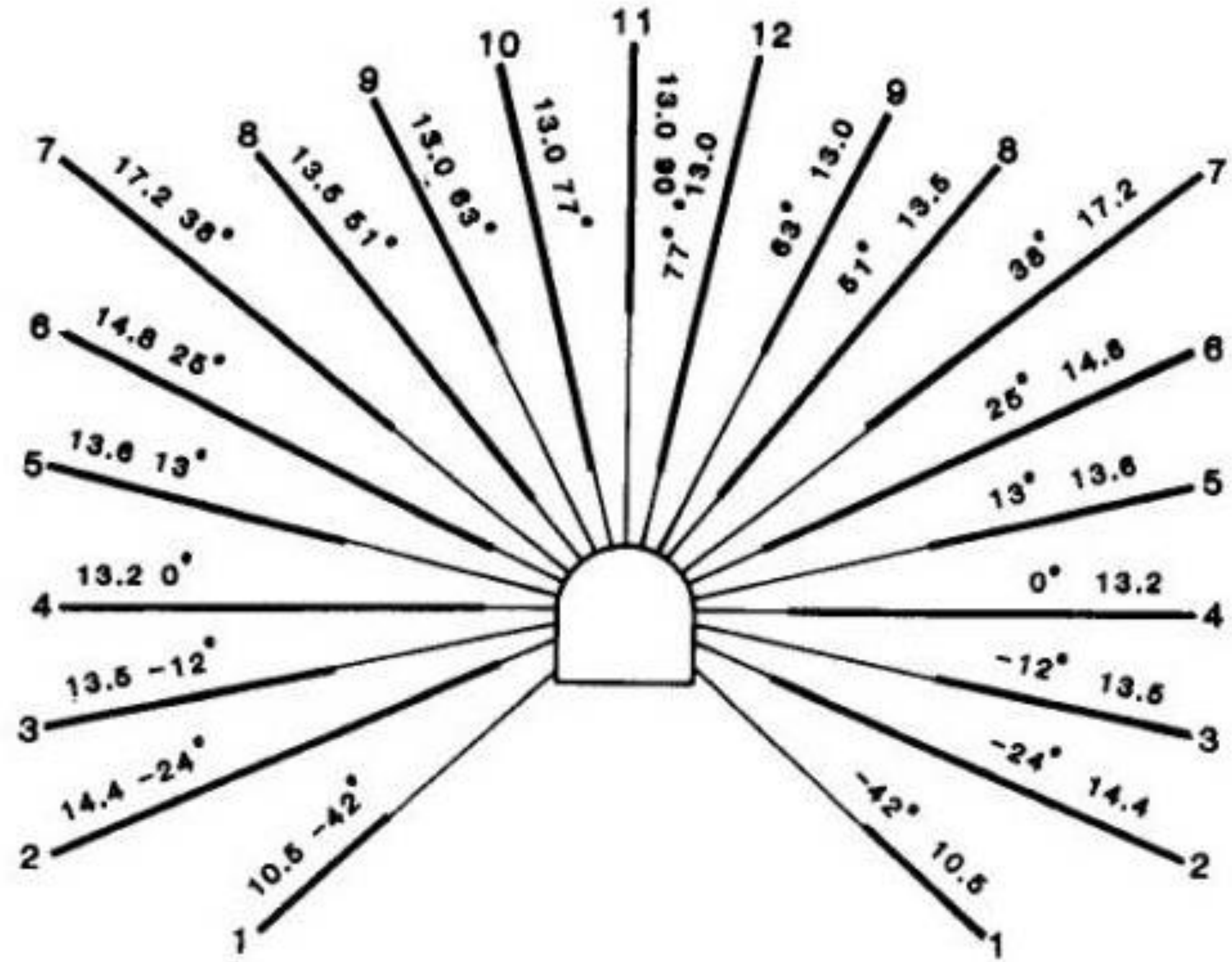


Fig. 20.3.15. Longhole drilling pattern.

Long hole drilling pattern for under cut

Draw Points I

*Draw points for grizzly & slusher system are driven at right angle to grizzly or slusher drift.

*Draw points are inclined, angle varies from vertical to 45 degree.

*Cross section of draw points are large enough to pass most of rock fragment with out requiring secondary blasting.

*After driving inclined section of draw point out of draw point centre, a vertical section be driven up wards to elevation of under cut level.

Draw Points II

*Vertical portion of ore pass raise under cut further above production level and also make pillar more stable.

*Concrete lined draw points allow better flow of broken rocks & retain its size for longer time.

*Concrete lined draw points can be repaired easily in compared to unlined draw points.

*Size of draw point should be maintained to prevent flooding of production drift with broken ore.

Draw Points III

*In LHD system, draw point entry is horizontal for allowing LHD to operate.

*Connection between production level & under cut level is a fairly large draw cone formed by drilling & blasting.

*Draw points designed to allow larger pieces to move down to brow of draw point.

Draw Points IV

*At this place big boulders are drilled & blasted, if boulders are big enough for LHD to handle.

*Brow of draw point should be high enough to LHD to load,.

*Very high brow shall allow broken rock to flood production drift, causing hindrance in LHD mucking.

Boundary Weakening I

*Boundary weakening is done to minimize side flow of waste material in to broken ore column.

*Boundary weakening facilitate caving by creating weaker boundary with waste rock, thus reducing dilution.

*All rock has tendency to arch at time of caving, hindering caving operation.

*Boundary weakening reduces arching, thus facilitating caving.

*Boundary weakling should be placed on two adjusent side of initial caving area.

Boundary Weakening II

*Extend of wreaking depends on on strength of ore.

*For very weak, highly fractured ore body, some corner raises and drilling along two boundaries may be sufficient.

*For stronger, less fractured ore body, a completely fractured zone about 2.5m to 3m may be required.

*This type of boundary weakening is formed by driving sub levels drift along boundaries at vertical interval of 18m.

*Number of sub levels will be determined by height of ore column & strength of rock.

Boundary Weakening III

*Two or three lines of vertical drill holes are then drilled between sublevels.

*These drill holes are loaded & blasted, to produce a completely fractured zone, that will weakening two legs of a potential arch & greatly facilitate initial caving action.

Boundary weakening by blasting

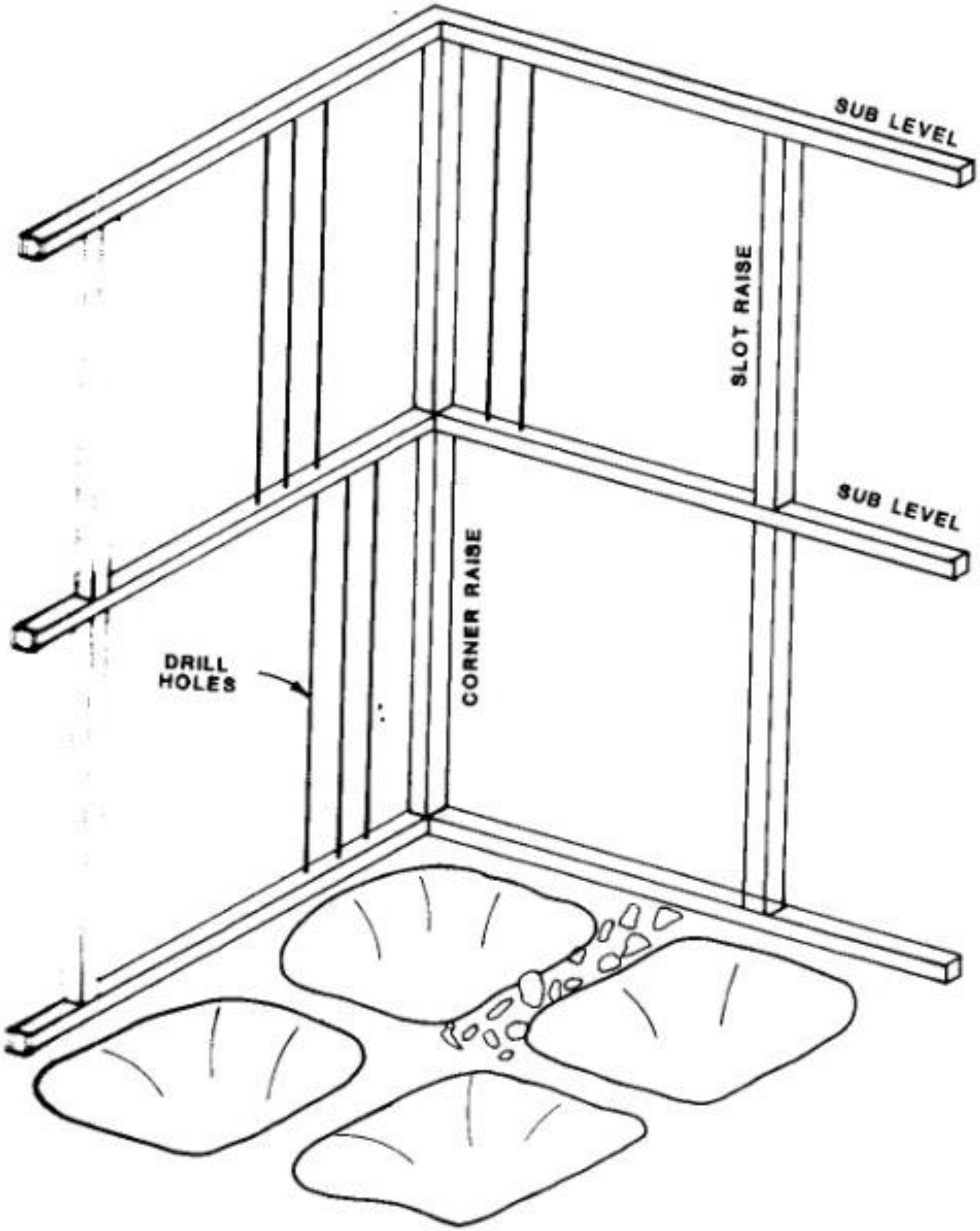


Fig. 20.3.5. Boundary weakening.

Under Cut Blasting I

*Under cut is last step to initiate caving.

*Under cutting is very critical operation & any failure in this operation shall critically effect caving/

*Few long holes are to be blasted at one time, so that checking of pillars can be done for taking corrective measures.

*All pillars should be removed before next blasting sequence.

*If pillars are left in cave area, they shall hinder caving action & transmit damaging stress in to working below.

Under Cut Blasting II

*Correcting or repairing of any under cutting activities is very costly.

*Some time, correction is not possible, thus loosing ore in stope.

Secondary Blasting I

*Caving action always produces some over size boulders that will not pass through draw points.

*It is specially true during initial caving period when attrition has not had an opportunity to reduce size of rock fragments.

*Common method of breaking oversize rock is by use of concussion blast (plaster Shooting).

*2 to 4 kg of explosive is placed in strategic spots between rock fragments with help of long stick.

Secondary Blasting II

*When hang up of several draw points are to be blasted simultaneously, then blasting cord will be tied together & ignited by 1 detonator.

*When hang up consists of cluster of smaller rocks, a high pressure water hose can be used to loosen hang-up.

*In LHD system, large rock fragments are usually close enough to draw points brow that can be drilled with carrier mounted jackleg or small single boom drill jumbo & loaded with explosive.

*Shooting hang-up with slusher system is hazards work & employees require special training.

Support I

*Support is very important in block caving.

*As cave area enlarged, there shall be redistribution of stress in surrounding area including under cut area.

*It is important to maintain as much as initial integrity of rock as possible.

*Rock bolt & concrete are most common method of support used in block caving method.

*Rock bolts should be installed as soon as possible.

*It is prefer to have rock bolting as an integral part normal drilling, blasting & mucking cycle.

*Main haulage drive, grizzly drift, slusher drift, LHD draw point to be supported by concrete,

Support II

*Some places, timber & steel sets have been used to support main haulage drive.

*Reinforced concrete is not recommended where blasting concussion will be present.

*It has been found that vibration / shock wave separated concrete form reinforcement.

*Loose bar hanging out in drift causes operational problem.

*Loss of reinforcement shall cause subsentoal loss of strength in support.

*Concrete should have high strength, no segregation & cold joints.

Ventilation I

*Strong, well monitored ventilation system is essential for block caving.

*Production place produces substacial amount of dust, especially if ore is dry.

*Contaminated air should be removed from work places as soon as possible through return air path.

*Fresh air should not get mixed up with contaminated air.

*Ventilation system should allow to send controlled amount of air to haulage drift, production drift & individual work places.

Ventilation II

*Amount of air circulated depends on number of work places being utilized at any given time.

*Air requirement shall also depends on type of machinery being used in mine, diesel equipment needs more air that compressed air operated machines.

*Work place volume to be ventilated is much less, in compare to other method of working, hence ventilation is generally effective.

Height of Block I

*Critical height of ore is such that mineral content will pay for development, production, milling, mining and over head cost.

*Grade of ore & selling price shall have a substacial effect on critical height.

*Productive life of each work place is considered for deciding critical height.

*Boundary weakening up to certain vertical length is practically effective.

*Boundary weakening limits maximum stope height.

Height of Block II

*Draw points, chutes, ore pass haulage level have certain life span based on time & ore to be handled.

*Higher column height increases passivity of dilution by mixing ore with waste rock.

*Beyond a certain height of ore column, mining shall not be possible.

Dilution Control I

*Dilution factor varies from 10 to 20% of total ore drawn.

*Permissible dilution level depends on ore grade, cost & metal price.

*Controlling dilution is very critical for block caving method.

*Draw must be distributed evenly over entire cave area.

*Rate of draw should not vary substantially between draw points.

Dilution Control II

*Ore adjacent to an uncaved block should not completely drawn down before adjacent ore has been caved.

*Ore adjacent to an uncaved block should not completely drawn down before adjacent ore has been caved.

*Dilution will easily migrate from sides into newly caved area, cutting off ore above & preventing it from caving.

*Good draw control is to be maintained to achieve minimum area of contact between ore & waste zone above.

Dilution Control III

*Concentrating production in small area of caved ore will create a funnel for dilution to flow down into ore zone to become intimately mix with surrounding ore.

*Dilution can also be controlled by increasing equal percentage increment in each line of draw points progress away from cave line.

*Principal is to maintain an ore to waste contact of 45 to 50 degree away from uncave ore.

Dilution control for wider stope by keeping cave profile at angle.

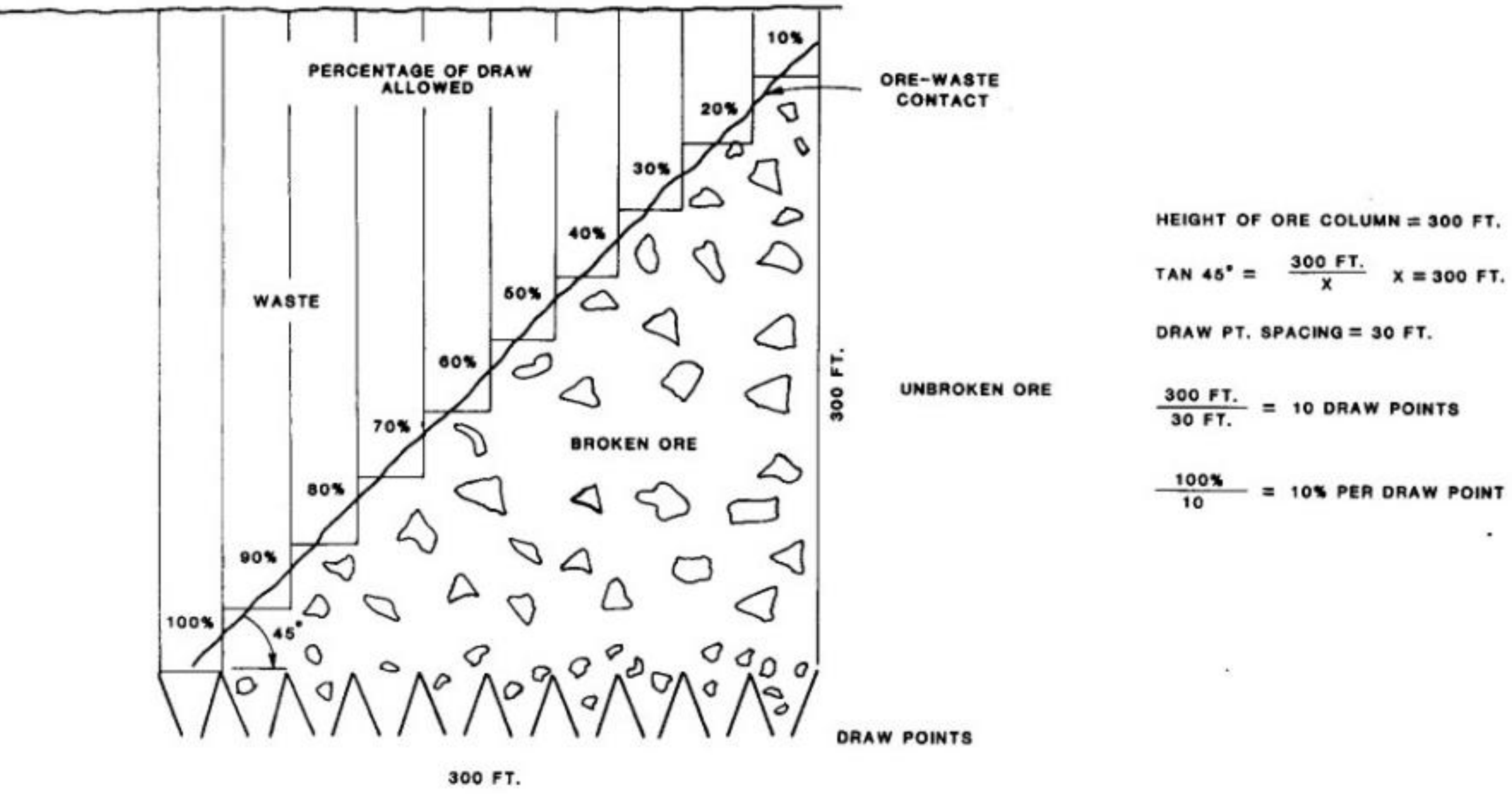


Fig. 20.3.6. Draw control section. Conversion factor: 1 ft = 0.3048 m.

Mining Limit I

*Mining limits are determined by economic limits in porphyry deposit.

*In some case, there is no definite separation between mineralized & waste rock.

*Grade of ore gradually diminishes until it become s to low to recover economically.

*Mining limit is set by minimum height of economic grade ore that will still contribute to profitability of mine.

*In case when lateral extend of ore body is very large, it will not be possible to mine full width as one block.

Mining Limit II

*In this case, mining levels are to be divided in several blocks and mine them sequentially.

*In case, there is two mining blocks, new mining block should be immediately adjacent to first mining block.

*In case, when mining is dine on two sides of an unmined block, potential rick pressure in unmined block becomes very high.

Caving I

*In caving, all rock has a natural tendency to form an arch .

*Dimension requires to produce an active cave depends on strength of rock mass.

*If rock mass is highly fractured & weak, a fully active cave will start with a fairly small under cut area.

*Stronger rock mass needs larger under cut area to establish an active cave.

*Mine layout should provide for expanding under cut area in two adjacent areas.

Caving II

*Minimum dimension of cave area will determine when an active cave will start.

*If an arch is formed over short axis of under cut area, then expanding cave along line of long axis will probably not produce a fully active cave.

*In order to break stable arch over under cut area, another line of under cut should be blasted, preferably along short axis.

*This will cause additional caving action until another stable arch is formed.

*Then another line of under cut should be blasted.

Caving III

*Eventually, shear strength of rock mass will be overcome and caving will be continuous.

*Boundary weakening can be great aid in establishing an active cave especially during initial caving sequence.

Induce Caving I

*Usually increasing size of undercut area will eventually produce a cave.

*There may be occation where under cut area becomes so large that it does not seem advisable to continue under cutting until an active caver is established.

*There may be a occasion, where mining limit has been reached in one direction & no farther expansion is possible.

*In this situation, caving can be continued by induced caving.

Induce Caving II

*Incase of new mine, with some excavation available from previous mining, old excavation can be used as access.

*In case mine has provision of observation drift, then this drift can be used for access above cave area.

*When non of this arrangements are available, some auxiliary raising and drifting may be necessary .

*Induced caving is done with help of long hole blasting.

*Long holes are drilled around perimeter of undercut area & blasted as pre split.

Induce Caving III

*This blasting induces further caving action.

*Long holes drilled & blasted over top of stable arch promotes some additional caving action.

*In case if drilling out over top of arch, efforts should first be made to locate top of arch so that long holes will be properly located & explosive charge can be calculated with some confidence.

*Some time a small drift is excavated over stable arch,

*Drift is loaded with explosive & blasted.

*Drift has to be stemmed to prevent gunbarrelling.

Cave Action I

*Observation drift along one or two sides of initial cave area at some distance above undercut level and out side cave line is excavated.

*Cross cuts are driven over to cave limit.

*In case open arch is forming, arch can be observed from cross cut.

*It is also possible to hear caving action if cave is active.

*In case arch became stable and caving action stopped, observation drift can be used to drill & blast long holes out over arch to help restart caving action.

Cave Action II

*This method is an expansive method.

*Once cave is fully active, observation points are covered with broken rock and no longer useful.

*drill holes can be used to install co-axial cables that are cemented in holes.

*As caving progresses up wards, cable will bend or break.

*Using galvanometer, resistance of cable can be measured to determine location of break.

*Kink or break may occur in cable where there is a displacement of rock.

Cave Action III

*This method does not locate top of broken ore, method is not accurate.

*Wooden or metal plug attached to a cable can be dropped down drill holes to measures top of arch.

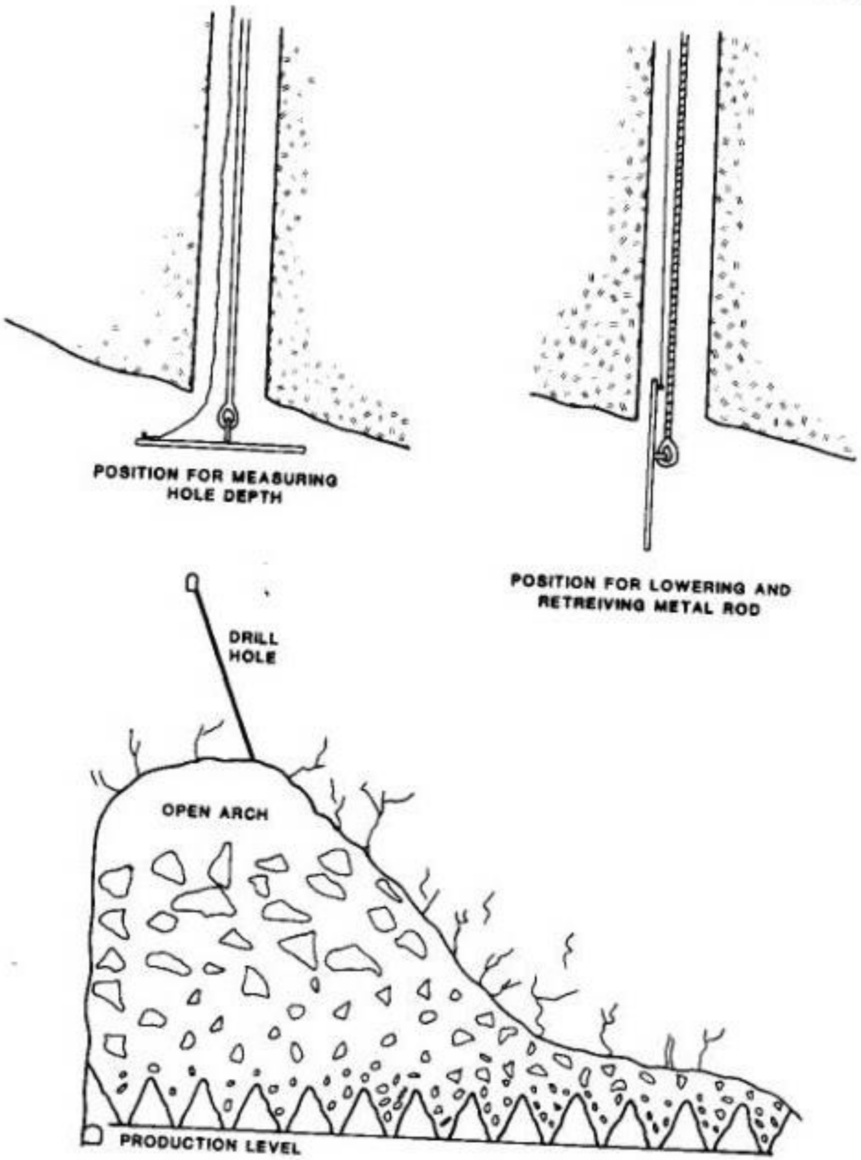
*When plug reaches an obstacle (broken ore) yhan plug is pulled back up hole until it reaches arch back.

*By measuring length of cable, top of aech can be measured fairly accurately.

*Top of broken ore is approximately located.

*More drill holes will provide better result.

Cave monitoring by bore hole



METHOD FOR MEASURING CAVE PROGRESS

Fig. 20.3.7. Cable and plug for finding top of arch.

Stress Problem I

*Highly jointed ore and matrix offer low resistance to stress

*Stress problem is inherent to block caving.

*As mine development progresses, left out pillars and surrounding area goes under stress.

*More development causes more stress on pillars and surrounding areas.

*Production area of block caving is subjected highest amount of pressure.

*Blasting of under cut releases sudden pressure yhat can cause cracking of concrete support.

Stress Problem II

*It is better to place rock bolts in concrete to prevent slabbing.

*By advancing under cut rapidly, stress duration for working can be reduced considerably.

*Production & undercut level development should nit be done much in advance, to reduce deterioration of developed area.

*In case, any un blasted pillar is left at under cut area, then remnant of pillar will not allow proper caving.

*This shall result sever stress concentration at production area.

Stress Problem III

*All working in critical are should be well supported.

*Once caving reaches surface, stress problem gets reduced.

*Cave line should never be stopped for an extended period of time immediately over or adjacent to main working.

*In this case, development shall be subjected to high stress for longer time.

*Extensometer should be installed at drive, cross cut to monitor stress problem.

Subsidence I

*Surface subsidence will always occur with block caving.

*Time from initial production to first surface disturbance will depend on depth of mining kevel below surface, strength of ore & rate of draw.

*First sign of subsidence is likely to be circular hole or funnel appearing somewhere with in boundaries of undercut area.

*Subsidence can also be a general settling of surface with in boundaries of undercut area.

*Total area of caving will settle & escarpment will form.

Subsidence II

*Escarpment may be nearly vertical to as flat as 45 degree or less, depending on rock strength & topography.

*Tension cracks will form around escarpments.

*If a 45 degree line is drawn from undercut level to surface, this is the area in which tention crecking may occur.

*No permanent structure should be placed inside this line since there is a high probability that will be disturbed by tension crack.

Reference I

*SME Mining Engineering Hand Book, 2nd Edition, Volume 2, page 1815 to 1836